INCH-POUND
MIL-DTL-24709A(SH)
21 February 2018
SUPERSEDING
MIL-D-24709(SH)
4 November 1988

DETAIL SPECIFICATION

PADS, DISTRIBUTED ISOLATION MATERIAL, TYPE R-50-21/64

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Navy type R-50-21/64 homogeneous rubber pads composed of distributed isolation material (DIM) for use in machinery and equipment mounts to isolate vibration and attenuate shock.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Military Marking for Shipment and Storage

MIL-STD-177 - Rubber Products, Terms for Visible Defects of

MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items

(Copies of these documents are available online at http://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D395 - Standard Test Methods for Rubber Property – Compression Set

ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension

ASTM D471 - Standard Test Method for Rubber Property – Effects of Liquids

ASTM D573 - Standard Test Method for Rubber – Deterioration in an Air Oven

ASTM D792 - Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement

ASTM D1149 - Standard Test Methods for Rubber Deterioration—Cracking in an Ozone Controlled Environment

ASTM D1229 - Standard Test Method for Rubber Property – Compression Set at Low Temperatures

ASTM D1418 - Standard Practice for Rubber and Rubber Latices – Nomenclature

ASTM D2240 - Standard Test Method for Rubber Property – Durometer Hardness

ASTM D3182 - Standard Practice for Rubber – Materials, Equipment, and Procedures for Mixing Standard Compounds and Preparing Standard Vulcanized Sheets

ASTM D3767 - Standard Practice for Rubber – Measurement of Dimensions

ASTM D5992 - Standard Guide for Dynamic Testing of Vulcanized Rubber and Rubber-Like Materials Using Vibratory Methods

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 1382 - Rubber – Vocabulary

ISO 10846-2 - Acoustics and Vibration – Laboratory Measurement of Vibro-Acoustic Transfer Properties of Resilient Elements – Part 2: Direct Method for Determination of the Dynamic Stiffness of Resilient Supports for Translatory Motion

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ISO 10846-3 - Acoustics and Vibration – Laboratory Measurement of Vibro-Acoustic Transfer Properties of Resilient Elements – Part 3: Indirect Method for Determination of the Dynamic Stiffness of Resilient Supports for Translatory Motion

(Copies of these documents are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material.

3.2.1 Composition. The pads shall be fabricated of oil-resistant compounds utilizing polymerized chloroprene (ASTM D1418 designation CR) or nitrile rubber (ASTM D1418 designation NBR) as the basic material. The use of any rubber that gives off toxic or damaging volatiles at operating temperatures up to 125 degrees Fahrenheit (°F) (52 degrees Celsius [°C]) shall be prohibited. No acceptable toxicity requirements exist for temperatures above 125 °F.

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Appearance. Rubber DIM pads shall be free of visual defects in accordance with table IV and MIL-STD-407 as specified in 4.4.2.

3.4 Dimensions.

3.4.1 Pad dimensions. Unless otherwise specified (see 6.2), the rubber DIM pad shall be 18 by 18±¼ inches (457 by 457±6 millimeters [mm]) excluding the proof buttons. The thickness of the rubber DIM pad shall be 0.328±0.016 inch (8.3±0.4 mm). Dimensions of the pad shall be measured in accordance with ASTM D3767 as specified in 4.4.2.1.

3.4.2 Hardness-proof buttons. The pads shall have two hardness-proof buttons molded onto opposite edges and lying outside the patterned portion of the pad, and shall be securely attached to the parent pad by flash or runners. The buttons shall be of nominal 1.25 inches in diameter by 21⁄64 inch thick (32 by 8.3 mm).

3.4.3 Surface pattern. Unless otherwise specified (see 6.2), dimensions of the surface pattern shall be according to the manufacturer's preference with a tolerance of ±0.010 inch (0.26 mm) (see 3.6.1). Dimensions and distributions of voids of the surface pattern shall not preclude cutting the DIM pad into annular (machinery-mounting) pads with walls as narrow as ½ inch (13 mm).

3.5 Properties of rubber stock and DIM pads. Rubber stock and DIM pads shall conform to the requirements shown in table I when tested as specified herein.

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TABLE I. Requirements for rubber stock and DIM pads.

Section	Properties		Requirements
3.2	Material		Polymerized chloroprene (ASTM D1418 designation CR) or nitrile rubber (ASTM D1418 designation NBR)
3.3, 3.4	Appearance and dimensions		18 by 18±¼ inches (457 by 457±6 mm); thickness of 0.328±0.016 inch (8.3±0.4 mm)
3.5	Rubber stock property	Initial tensile strength	1,800 pounds per square inch (psi) (12.4 Megapascals [MPa])
		Elongation	350%
3.5	Rubber stock property	Tensile strength	1,700 psi (11.7 MPa)
		Elongation after oven age	300%
3.5	Rubber stock property	Tensile strength	1,700 psi (11.7 MPa)
		Elongation after oil immersion	300%
3.5	Rubber stock property: volume change after oil immersion		±10%
3.5	Rubber stock property: cold compression set after oil immersion		Maximum 35%
3.6.1	DIM pad property: deflection at load		0.065 inch (1.65 mm) maximum deflection; height no less than 0.247 inch (6.27 mm)
3.6.2	DIM pad property: resonant frequency before compressive strength test		27 hertz (Hz) at 25 psi (0.172 MPa) 22 Hz at 50 psi (0.344 MPa)
3.6.3	DIM pad property: compressive strength		Shall not rupture at 10,000 psi test load
3.6.4	DIM pad property: resonant frequency after compressive strength test		22±1 Hz at 50 psi
3.6.5	DIM pad property: drift		Not to exceed 8 mils (0.20 mm)
3.6.6	DIM pad property: hardness		First article value ±5 shore A hardness units
3.6.7	DIM pad property: specific gravity		First article value ±0.02
3.6.8	DIM pad property: ozone resistance		Shall not crack under test
3.7	Drawings		As required (see 6.2)
3.8	ID markings		In accordance with MIL-STD-129
3.9	DIM material: toxicity		Shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose

3.6 Performance requirements.

3.6.1 Deflection at rated load. The deflection of the DIM pads at the 50 lb/in^2 (0.345 MPa) load point shall not exceed 0.065 inch (1.65 mm) when tested as specified in 4.4.4.1. Compressed height at the 50 lb/in^2 (0.345 MPa) load point shall be not less than 0.247 inch (6.27 mm) minimum. If present, the surface pattern height shall not be included in the measurement of deflection.

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3.6.2 Resonant frequency before compressive strength test. The resonant frequency of vibration of the rubber pad with the supported load shall not exceed the values specified in table II when measured in accordance with the procedures of ASTM D5992, ISO 10846-2, or ISO 10846-3, as specified in 4.4.4.2.

TABLE II. Resonant frequency requirement.

Load (lb/in^2)	Resonant frequency (Hz)
25 (0.172 MPa)	27.0
50 (0.345 MPa)	22.0

3.6.3 Compressive strength. The pad shall not rupture when subjected to the compressive strength test specified in 4.4.4.3.

3.6.4 Resonant frequency after compressive strength test. After being subjected to the compressive strength test (see 3.6.3), the resonant frequency of the rubber pad shall be changed by not more than ±1 Hz when retested in accordance with ASTM D5992, ISO 10846-2, or ISO 10846-3, at a supported load of 50 lb/in^2 (0.345 MPa) as specified in 4.4.4.4.

3.6.4.1 Methods for determining resonant frequency. ASTM D5992, ISO 10846-2, or ISO 10846-3 measurement methods for determining resonant frequency of pad samples under specified loads shall be used. All methods listed above are acceptable and will provide comparable results.

3.6.5 Drift. The drift of the rubber pad shall not exceed 8 mils (0.20 mm) when tested in accordance with 4.4.4.5.

3.6.6 Hardness. The hardness of the pads shall be determined by the procedure specified in 4.4.5. The hardness value found during first article inspection (see 3.1) of a contractor's DIM pads shall establish the initial requirement with a tolerance of ±5 shore A hardness units for subsequent conformance testing of the contractor's pads.

3.6.7 Specific gravity. The specific gravity of the rubber in the DIM pad shall be determined by the procedure specified in 4.4.6. The specific gravity found during first article inspection (see 3.1) of the contractor's DIM pads shall be recorded as a requirement with a tolerance of ±0.02 for subsequent conformance testing of the contractor's pads.

3.6.8 Ozone resistance. The rubber pad shall not be cracked by exposure to ozonized air when tested in accordance with 4.4.7 during first article inspection.

3.7 Drawings. (See 6.2 and 6.3)

3.8 Identification marking. DIM pads shall be identified by markings in accordance with MIL-STD-129.

3.9 Prohibited materials. The DIM rubber pads shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

3.9.1 Material safety data sheet (MSDS). When specified (see 6.2 and 6.5), the MSDS shall be in accordance with FED-STD-313.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 First article inspection. First article inspection shall consist of the verifications specified in table III. Should any of the pads submitted for first article inspection exhibit major or minor defects as specified in 4.4.2 and 4.4.3, or if the pads fail the tests as specified in 4.4.4, this shall be cause for rejection.

4.2.1 Sampling for first article. The following samples shall be selected for first article inspection:

a. Three sheets of the rubber stock, 6 by 6 by 0.080±0.010 inches (150 by 150 by 2 mm), shall be prepared in accordance with ASTM D3182. The sheets shall be made from the same rubber compound as the rubber pads with an equivalent cure. Grain direction shall be marked to facilitate tension testing.

b. Three rubber DIM pads with attached hardness-proof buttons representative of production under the contractor's identifying number and as specified in 3.2.

c. Four compression set buttons, 1.14±0.02 inches (29.0±0.5 mm) in diameter with a thickness of 0.50±0.02 inch (13.0±0.2 mm), prepared in accordance with ASTM D395. The buttons shall be made from the same rubber compound as the rubber pads with an equivalent cure. Buttons need not have pad surface design features.

4.3 Conformance inspection. Conformance inspection shall consist of the examination, measurements, and tests specified in table III. Conformance samples shall be as specified in 4.2.1.

TABLE III. First article and conformance inspections.

Requirement	Verification	First article			Conformance		
		Stock	Button	Pad	Stock	Button	Pad
3.2	4.4.2	X	---	---	X	---	---
3.3, 3.4	4.4.2	---	---	X	---	---	X
3.5	4.4.3.1	X	X	---	X	X	---
3.5	4.4.3.2	X	X	---	---	---	---
3.5	4.4.3.3	X	X	---	---	---	---
3.5	4.4.3.4	X	X	---	---	---	---
3.5	4.4.3.5	X	X	---	---	---	---
3.6.1	4.4.4.1	---	---	X	---	---	X
3.6.2	4.4.4.2	---	---	X	---	---	X
3.6.3	4.4.4.3	---	---	X	---	---	---
3.6.4	4.4.4.4	---	---	X	---	---	---
3.6.5	4.4.4.5	---	---	X	---	---	---
3.6.6	4.4.5	---	X	---	---	X	---
3.6.7	4.4.6	X	---	---	X	---	---
3.6.8	4.4.7	---	---	X	---	---	---
3.8	4.4.2	---	---	X	---	---	X
3.9	4.5	X	---	---	---	---	---

4.3.1 Lot. A lot for conformance inspection shall consist of all rubber DIM pads produced in one plant under essentially the same conditions and offered for delivery at one time. In each case the number of DIM pads shall be the lot size.

4.4 Procedures for examination, measurements, and tests.

4.4.1 Temperature of tests. Unless otherwise specified (see 6.2), every sample shall be conditioned for at least 4 hours at 80±9 °F (26.7±5 °C) before being tested, and the tests shall be performed at this temperature.

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4.4.2 Visual and dimensional examination. Each of the rubber DIM pads submitted for first article inspection (see 4.2.1) or sampled for conformance inspection (see 4.3.1) shall be examined for major and minor defects in accordance with MIL-STD-407 to determine conformance to 3.3 and 3.8. Surfaces of the DIM pad shall be classified as location I in accordance with MIL-STD-407 for the purposes of this examination. Bloom of the surface of the pad shall not be considered a defect (see table IV).

4.4.2.1 Dimensions. The length, width, and thickness of the DIM pad and the dimensions of the surface design shall be measured in accordance with ASTM D3767. This shall be done to define the surface design on the DIM pad during first article inspection and to determine conformance to 3.4. During conformance measurements, any deviation of the surface design defined for first article (see 3.1), after allowing for the 0.010-inch (0.26-mm) tolerance (see 3.4), shall be regarded as a major defect as shown in table IV to determine acceptance or rejection. Any deviation of the length, width, and thickness of the DIM pad from the specified dimensions, after allowing for tolerances (see 3.4 and 6.2), shall be regarded as a major defect.

4.4.3 Tests on special rubber samples. The properties of the 0.080-inch (2-mm) thick samples and compression set buttons (see 4.2.1) representing the rubber pads shall be evaluated to determine conformance to the requirements of 3.5.

4.4.3.1 Initial tensile strength and ultimate elongation. The initial tensile strength and ultimate elongation of the rubber stock shall be determined in accordance with ASTM D412 using Die C samples.

4.4.3.2 Tensile strength and ultimate elongation after oven aging. The tensile strength and ultimate elongation of the rubber stock shall be determined after oven aging in accordance with ASTM D412 and ASTM D573 except that the aging period shall be 70±¼ hours at a temperature of 212±2 °F (100±1 °C). Die C samples shall be used.

4.4.3.3 Tensile strength and ultimate elongation after oil immersion. The tensile strength and ultimate elongation of the rubber shall be determined after immersion in ASTM No. 1 oil for 46±¼ hours at a temperature of 158±2 °F (70±1 °C) in accordance with ASTM D412 and ASTM D471 using Die C samples. The tensile strength reported shall be based on the original cross section of the samples.

4.4.3.4 Volume change after oil immersion. The change in volume of the rubber shall be determined after immersion in ASTM No. 1 oil for 46±¼ hours at a temperature of 158±2 °F (70±1 °C) in accordance with ASTM D471.

4.4.3.5 Cold compression set. Cold compression set shall be determined in accordance with ASTM D1229, except that the exposure shall be 30±2 °F (minus 1±1 °C) for 94±½ hours. Compression set shall be determined 30 minutes after release from the jig.

4.4.4 Destructive tests on pads.

4.4.4.1 Deflection at rated load. Three samples, each 2¾±1/16 inches (70 mm) in diameter with a centrally-located hole, 1⅛±1/16 inches (28.6 mm) in diameter, shall be prepared from a DIM pad. The samples shall be cut from the pads by means of a precision cutting tool (such as a waterjet, laser, flash CNC cutter, die, or rotating cutting tool). They shall be lubricated as necessary such that a smooth-cut surface having square edges is obtained. The exact projected surface area of each sample shall be calculated from measurement of these dimensions to the nearest 1/64 inch (0.4 mm).

a. The deflection under load of each sample in turn shall be determined using a universal testing machine capable of measuring the load in units not larger than 5 pounds (22 newtons) and deflection in units not larger than 0.005 inch (0.13 mm). Each sample shall be subjected to four loading and unloading cycles up to a peak load of 75 lb/in^2 (0.517 MPa). For the first three cycles, the deflection rate shall not exceed 0.2 inch per minute (in/min) (5 millimeters per minute [mm/min]). For the fourth cycle, the rate of loading and unloading shall not exceed 0.05 in/min (1.3 mm/min). The deflection of the sample in inches measured during the loading phase of the fourth cycle shall be plotted versus load, and the deflection produced by a load of 50 lb/in^2 (0.345 MPa) shall be read from the curve. The average of the values obtained on the three samples at this load shall be used to determine conformance to 3.6.1.

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b. The loaded height of each of the three samples at 50 lb/in^2 (0.345 MPa) load shall be measured. This measurement shall be made at the same time as the deflection-load test (see 4.4.4.1.a), using a dial gauge graduated in thousandths of an inch, and shall be taken during the loading phase of the fourth cycle. The three values obtained shall be averaged to determine conformance with the loaded height criteria of 3.6.1.

4.4.4.2 Resonant frequency before compressive strength test. The same three samples used for the deflection-under-load test (see 4.4.4.1.a) shall be used for the resonant frequency test. The resonant frequencies of vibration of each sample shall be determined by one of the methods of ASTM D5992, ISO 10846-2, or ISO 10846-3, while supporting purely mass loads of 25 lb/in^2 (0.172 MPa) and 50 lb/in^2 (0.345 MPa), respectively. The clean, dry, test sample shall be compressed between clean, dry, polished steel platens. The vibration excursion (double amplitude) shall not exceed 0.01 inch (0.25 mm) during the test. The averages of the resonant frequency values obtained for the three samples at each load shall be used to determine conformance to 3.6.2.

a. Larger DIM pads having patterns not previously tested. An additional sample, 3¾±1/16 inches (95 mm) in diameter with a centrally-located, 1⅛±1/16 inches (29 mm) hole, shall be tested for resonant frequency at 25 lb/in^2 (0.172 MPa). Results of this check shall also be within the limit specified in 3.6.2.

b. Calibration of resonant frequency test systems. Any system used in the resonant and non-resonant testing of pads to determine the resonant frequency shall be calibrated before use. Steel coil springs of known spring rate may be used for calibration purposes to ensure that the equipment behaves correctly with changes in stiffness, and in a linear manner with variations of excitation amplitude and frequency. Care shall be taken to ensure that the compliance of the test equipment does not result in erroneously low values for resonant frequency and that the error due to this effect does not exceed 1 percent of the correct resonant frequency value. The test equipment should be at least a factor of 100 times stiffer than the sample being tested.

4.4.4.3 Compressive strength. Each of the three 2¾-inch (70-mm) samples tested in 4.4.4.2 shall be subjected to a load of 10,000 lb/in^2 (69 MPa), one time, in a universal testing machine. The rate of loading and unloading shall not exceed 0.05 in/min (1.3 mm/min) with no dwell time at maximum load. Maximum deflection shall be observed and the samples examined to determine conformance to 3.6.3.

4.4.4.4 Resonant frequency after compressive strength test. The resonant frequency of vibration of each of the three samples subjected to the compressive strength test of 4.4.4.3 shall be determined while they support a load of 50 lb/in^2 (0.345 MPa). The test conditions specified in 4.4.4.2 shall be observed. The average of the resonant frequency values obtained for three specimens shall be used to determine conformance to 3.6.4.

4.4.4.5 Drift. One sample, 2¾±1/16 inches (70 mm) in diameter with a centrally-located hole 1⅛±1/16 inches (29 mm) in diameter, shall be prepared from the DIM pad. The exact surface of the sample shall be calculated from measurement of these dimensions to the nearest 1/64 inch (0.4 mm). The clean, dry sample shall be loaded in compression to 50 lb/in^2 (0.345 MPa) between clean, dry, polished steel platens using a deadweight. The height of the sample in mils shall be measured 1 hour after application of the load and 336 hours (2 weeks) after application of the load. The difference in the two readings in mils shall be reported as drift and used to determine conformance to 3.6.5.

4.4.5 Hardness test. Hardness of the DIM pads shall be determined on hardness proof buttons, 1.25 inches in diameter by 21/64 inch thick (32 by 8.3 mm), molded on opposite edges of the pad as specified in 3.4. Measurements shall be made in accordance with ASTM D2240 using a shore A durometer and 3-second readings. A single hardness value consisting of the average of the readings taken shall be obtained for each button. These values, two for each pad tested, shall be used to determine conformance to 3.6.6. Failure of either hardness button to conform to 3.6.6 shall be regarded as a major defect in determining acceptance or rejection for conformance (see 4.3) or during first article inspection (see 4.2).

4.4.6 Specific gravity. The specific gravity of the rubber compound stock materials used for the DIM pads shall be determined on samples cut from 6 by 6 by 0.080-inch sheets using ASTM D792. The average of three determinations shall be used to determine conformance with 3.6.7.

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4.4.7 Ozone resistance. Three samples, each 2¾±1/16 inches (70 mm) in diameter with a centrally-located, 1⅛±1/16-inch (29-mm) diameter hole, shall be prepared from finished DIM pads. The clean, dry samples shall be compressed between clean, dry, steel platens and held at a deflection corresponding to a load of 50 lb/in^2 (0.345 MPa) (see 4.4.4.1). The compressed samples shall be conditioned for 16 to 18 hours at a temperature of 100±2 °F (38±1 °C) in an oven in accordance with ASTM D573. The compressed samples shall then be exposed to air at a temperature of 100±2 °F (38±1 °C) containing ozone at a concentration of 1.00±0.05 parts per million by volume. The time of exposure shall be 166±½ hours. The exposure shall be carried out in a chamber conforming to ASTM D1149 and the measurement of the ozone concentration shall be in accordance with ASTM D1149. After exposure, the edges of the compressed samples shall be examined for cracks in the rubber to determine conformance to 3.6.8. A magnifying lens of at least 7X power shall be used for the examination.

4.5 Prohibited materials. The absence of prohibited materials shall be verified as specified (see 3.9 and 6.2).

TABLE IV. Summary of visible rubber defects.

MIL-STD-407 [1/]	Description [2/, 3/]	Major	Minor
Air check (surface marking and depression due to trapped air in curing [MIL-STD-177])	Location I	X	---
	Location II	---	X
Blister (defect in a rubber product evidenced by a cavity or sac that deforms the surface [ISO 1382])	Extending through material body, location I or II	X	---
	Not extending through the material body, location I	X	---
	Not extending through the material body, location II	---	X
Bloom (liquid or solid material which has migrated to the surface of a rubber product. Note: Bloom changes the surface appearance of the rubber [ISO 1382]).	Heavy, location I	X	---
	Heavy, location II	---	X
	Light, location I	---	X
	Light, location II [4/]	---	---
Crack (fissured surface condition [MIL-STD-177])	Extending through the material body, location I or II	X	---
	Not extending through the material body, location I	X	---
	Not extending through the material body, location II	---	X
Deformed (any defective change of form or shape [MIL-STD-177])	Location I or II	X	---
	Location II (if not affecting function)	---	X
Flash (excess material protruding from the surface of a molded product at the mold junction [ISO 1382])	Cured in, delamination, location I or II	X	---
	Cured in, no delamination, location I	X	---
	Cured in, no delamination, location II	---	X
Flow mark (marks or lines on a molded product, caused by imperfect fusion of flowing fronts [ISO 1382])	No internal delamination, location I	X	---
	No internal delamination, location II	---	X
	Any internal delamination	X	---
Foreign material (not defined)	Extending into the body, location I or II	X	---
	On surface, location I	X	---
	On surface, location II	---	X
Hole (material missing or void through body [MIL-STD-407])	Location I or II	X	---
Hole, pinhole (minute circular void or solvent blow hole [MIL-STD-177])	Individual pinhole, location I	X	---
	Individual pinhole, location II	---	X

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TABLE IV. Summary of visible rubber defects – Continued.

Porosity (presence of numerous minute voids in cured material [MIL-STD-177])	Extending into the body, location I or II	X	---
	Surface only, location I	X	---
	Surface only, location II	---	X
Sprue mark (excess material protruding from the surface of a molded product at the mold channels [ISO 1382])	Excess, location I or II	X	---
	Slight, location I	X	---
	Slight, location II	Not a defect	
Trim (not defined)	Material missing, location I or II	X	---
	Small amount of material missing, location I	X	---
	Small amount of material missing, location II	---	X
	Excess material (flash), location I	X	---
Trim (not defined)	Excess material (flash), location II	---	X

NOTES:

1/ This table omits these non-applicable defects which are in MIL-STD-407: cavity, coating, fill, identification not legible, dirty mold, register, and rind.

2/ Location I defect. The condition where the imperfection occurs on a primary use part of the item surface with regards to dimensional tolerances or physical stress in service (e.g., pad working surface).

3/ Location II defect. The condition where the imperfection does not occur on a primary use part of the item surface with regards to dimensional tolerances or physical stress in service (e.g., pad edges, depending on application).

4/ Where ozone resistance is required, light bloom necessarily results, and in this case is not a defect.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The oil and ozone-resistant DIM pad covered by this specification is intended for use in naval shipboard machinery and equipment mounts in order to isolate vibration and to attenuate shock, where temperatures do not exceed 125 °F (52 °C). Existing stock of DIM pads manufactured to MIL-D-24709 dated 4 November 1988 is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article is required (see 3.1 and 6.4).

c. Pad dimensions, when other than specified (see 3.4.1 and 4.4.2.1).

d. Surface pattern, when not at the option of the manufacturer (see 3.4.3).

e. Data required (see table I, 3.7, and 6.3).

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f. Requirement for verifying prohibited materials are not present (see 3.9 and 4.5).

g. When MSDS is required (see 3.9.3 and 6.5).

h. Temperature of tests, when other than specified (see 4.4.1).

i. Packaging requirements (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract: engineering drawings and their associated lists, first article inspection reports, and other inspection and test reports. The first article inspection report should include a detailed description of the test equipment and procedure used to determine resonant frequency (see 4.4.4.4.a).

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from among the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

6.5 Material safety data sheet (MSDS). Contracting officers should identify those activities requiring copies of MSDSs. Additional required Government information is contained in FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 should be in the contract.

6.6 Subject term (key word) listing.

Compressive strength

Hardness

Pad samples

Resonant methods

Specific gravity

Spring rate

Stiffness

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 5330-2018-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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